

Building Terrestrial Planets: The Four Dynamical Stages (Morbidelli et al. 2012)

From Runaway Coagulation to the Grand Tack, Water Delivery, and Giant Impacts

Stage 1: Runaway & Oligarchic Growth

Timeline: $t < 1$ Myr

- Dust grains coagulate into 1 – 100 km planetesimals via streaming instability.
- Gravitational runaway ($dM/dt \propto M^{4/3}$) transitions to oligarchic growth.
- Embryos grow to isolation mass $M_{\text{iso}} \approx 0.05 - 0.1 M_{\oplus}$ spaced by $\approx 10 r_H$.
- Inner disk (< 1 AU) is dry; water-ice condensates form beyond snow line (> 2.5 AU).

Embryo Swarm ($M \sim M_{\text{Moon}} - M_{\text{Mars}}$)



Stage 2: Jupiter & Saturn Grand Tack Migration

Timeline: 1 - 3 Myr (Gas Disk Phase)

- Jupiter undergoes Type II migration inward to ≈ 1.5 AU, truncating the inner disk at 0.95 AU.
- Saturn catches up, locking into mutual 3:2 Mean Motion Resonance.
- Common gap hydrodynamics reverses torques: Jupiter and Saturn migrate outward to > 5.2 AU.
- Mars feeding zone (1.2 – 1.8 AU) is severely depleted; Mars remains a stranded embryo ($0.1 M_{\oplus}$).

Inner Disk Truncated at 0.95 AU -- Resolves Mars Problem



Stage 3: Water & Volatile Delivery from Outer Belt

Timeline: 3 - 10 Myr

- Outward migration of giant planets scatters outer C-type carbonaceous embryos inward.
- Water-rich planetesimals (5 – 10% H_2O by mass) collide with growing proto-Earth and proto-Venus.
- Delivers 2 – 5 Earth oceans of water, matching terrestrial D/H isotopic chondritic signature.
- Asteroid belt is populated by a mixture of dry inner S-types and wet outer C-types.

Delivers ~ 3.5 Oceans ($M_{\text{water}} \sim 10^{-3} M_{\oplus}$)



Stage 4: Giant Impacts & Modern Architecture

Timeline: 30 - 100 Myr

- Gas disk dissipates ($\tau_{\text{gas}} \approx 3 - 5$ Myr), removing aerodynamic damping.
- Gravitational chaos triggers giant collisions among remaining $\sim 20 - 30$ embryos over 100 Myr.
- Moon-forming giant impact ('Theia' collision onto proto-Earth) occurs at $t \approx 50 - 100$ Myr.
- Final system settles into 4 planets: Mercury ($0.055 M_{\oplus}$), Venus ($0.815 M_{\oplus}$), Earth ($1.000 M_{\oplus}$), Mars ($0.107 M_{\oplus}$).

Final State: Low AMD ($S_d \approx 0.0018$), High RMC ($S_c \approx 89.9$)